



AIMS

African Institute for
Mathematical Sciences
RWANDA

Lecture 3: Probability III

Introduction to Probability and Statistics

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- 1 Random variables
- 2 Distribution of discrete r.v.'s
- 3 Distribution of Continuous r.v.

Random variables

A **random variable** (r.v.) X is a mapping that assigns a real number to each outcome:

$$\begin{aligned} X : \Omega &\longrightarrow V \\ \omega &\longmapsto x = X(\omega) \end{aligned}$$

A random variable can be

- **discrete**, that is $V \subseteq \mathbb{N}$. (ex : $\{0, 1\}$, $\{1, 2, 3, 4\}$, $\mathbb{N}$, etc.)
- **continuous**, that is $V \subseteq \mathbb{R}$ (ex : $[0, 1]$ ou $\mathbb{R}$, etc.).

Examples of random variable

- **discrete:**

- number of failures of a system component in a given time period,
- number of $\{Head\}$ obtained after flipping a coin n times,
- number of dots shown after rolling two six-sided dice.

- **continue:**

- times to failure of a device,
- necessary time to perform a given task,
- weights, heights of individuals in a population.
- measurement errors.

Examples of random variable

- Random experiment: Tossing a coin.
- Possible outcomes: *Tail* or *Head*.

- Universe:

$$\Omega = \{Tail, Head\}$$

- Associated random variable: $X(Tail) = 0$ et $X(Head) = 1$.
- $V = \{0, 1\}$.
- $Tail = X^{-1}(0)$ and $Head = X^{-1}(1)$.

Examples of random variables

- Random experiment: Rolling two dice simultaneously.
- Universe:

$$\Omega = \left\{ \left(\begin{array}{|c|c|} \hline \bullet & \bullet \\ \hline \end{array}, \begin{array}{|c|c|} \hline \bullet & \bullet \\ \hline \end{array} \right), \left(\begin{array}{|c|c|} \hline \bullet & \bullet \\ \hline \end{array}, \begin{array}{|c|c|} \hline \bullet & \bullet \\ \hline \end{array} \right), \dots, \left(\begin{array}{|c|c|} \hline \bullet & \bullet \\ \hline \end{array}, \begin{array}{|c|c|} \hline \bullet & \bullet \\ \hline \end{array} \right), \dots, \left(\begin{array}{|c|c|} \hline \bullet & \bullet \\ \hline \end{array}, \begin{array}{|c|c|} \hline \bullet & \bullet \\ \hline \end{array} \right) \right\}$$

- Random variable: X , sum of the number of dots on the face of the two dice.
 - For Example:

$$X \left(\begin{array}{|c|c|} \hline \bullet & \bullet \\ \hline \end{array}, \begin{array}{|c|c|} \hline \bullet & \bullet \\ \hline \end{array} \right) = 4.$$

- $V = \{2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$.

Measuring events

- r.v. versus event

$$\begin{aligned} X^{-1}(4) &= \{\omega \in \Omega \mid X(\omega) = 4\} \\ &= \left\{ (\boxed{\bullet}, \boxed{\bullet\bullet}), (\boxed{\bullet}, \boxed{\bullet\bullet}), (\boxed{\bullet\bullet}, \boxed{\bullet}) \right\}. \end{aligned}$$

- In practise, we denote:

$$P\{X^{-1}(4)\} = P\{X(\omega) = 4\} = P(X = 4),$$

- $P(X = x), x \in V$ is called **the distribution of X** or **the law of X** .

Distribution

Exercise

Toss an unfair coin (that is $P(H) \neq P(T)$).

What is the distribution of X the r.v. associated to the outcomes?

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What is the distribution of X the r.v. associated to the outcomes?

- $P(H) = P(X = 1) = p.$
- $P(T) = P(X = 0) = 1 - p.$

$$x \in \{0, 1\}, P(X = x) = (1 - p)^{1-x} p^x.$$

Distribution of a random variable

Exercise

What is the distribution of the r.v. X , sum of the dots obtained when rolling two dice simultaneously ?

Distribution of a random variable

Description of Ω

		Die 1					
							
Die 2		2	3	4	5	6	7
		3	4	5	6	7	8
		4	5	6	7	8	9
		5	6	7	8	9	10
		6	7	8	9	10	11
		7	8	9	10	11	12

Distribution of a random variable

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Die 1

						
Die 2	 2	3	4	 5	6	7
	3	4	5	6	7	8
	4	5	6	7	8	9
	 5	6	7	8	9	10
	6	7	8	9	10	11
	7	8	9	10	11	 12

Distribution of a random variable

The size of Ω is 36.

To compute $P(X = x)$, we count the number of couples such that the summation of the two dice dots is equal to x and we divide by $|\Omega| = 36$.

For example, $P(X = 5) = \frac{4}{36} = \frac{1}{9}$.

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Thus, the distribution of X is:

x	2	3	4	5	6
$P(X = x)$	1/36	1/18	1/12	1/9	5/36

x	7	8	9	10	11	12
$P(X = x)$	1/6	5/36	1/9	1/12	1/18	1/36



Distribution of a discrete r.v.

When a random variable is discrete, its law or distribution is described by the quantities $P(X = x)$, $x \in V$.

We have:

$$P(X = x) \in [0, 1], \forall x \in V \text{ et } \sum_{x \in V} P(X = x) = 1.$$

Distribution of a discrete r.v.

Exercise

Let us consider a population with a large number of individuals that may possess the character A .

Let p be the proportion of individuals that possess A .

We draw a sample of size n from the whole population.

Consider the random variable X , number of individuals in the sample possessing A .

What is the distribution of X ?

Hint: Consider the r.v. $Y_i = 1$ if the individual i has the character A , 0 otherwise and find the relationship between X and Y_i , $i = 1, \dots, n$.

Distribution of a discrete r.v.

Solution

Consider a r.v. Y_i such that $Y_i = 1$ if the individual i from the sample possesses the character A and $Y_i = 0$ otherwise.

From the text, $P(Y_i = 1) = p$ and $P(Y_i = 0) = 1 - p$.

$$\begin{aligned}\{X=x\} &= \{ \text{Amongst the } n \text{ individuals exactly } x \text{ possess the character } A \} \\ &= \{ x \text{ r.v. } Y_i \text{ are equal to } 1 \text{ and } (n-x) \text{ r.v. are equal to } 0 \}.\end{aligned}$$

There are several possible sequences of 0,1 containing exactly x times 1 and $(n-x)$ times 0.

Counting these sequences, we obtain:
$$C_n^x = \frac{n!}{x!(n-x)!}$$

(Write, count and compare the sequences with $n = 5$ and $x = 3$).

Continuous random variables

- X : Lap time for a given operation
- $\Omega = \mathbb{R}^+$.
- A possible outcome is: 1h, 27 min., 30 sec., 5/10 msec., ...
- In this case, it is not possible to associate an **exact** value to the outcome.
- Event: $X^{-1}(]a, b]) = \{\omega \in \Omega \mid a < X(\omega) \leq b\}$.
- The distribution is characterized by:

$$P\{X^{-1}(]a, b])\} = P\{a < X(\omega) \leq b\} = P\{a < X \leq b\}.$$

Distribution function

- When a random variable is continuous, we consider:

$$P(x < X \leq x + h) = P(X \leq x + h) - P(X \leq x).$$

We define the **cumulative distribution function** (c.d.f.).

Definition

The **cumulative distribution function** is the function $F_X : \mathbb{R} \rightarrow [0, 1]$ defined by

$$F_X(x) = P(X \leq x).$$

Cumulative distribution functions

Definition

A function F mapping the real line to $[0, 1]$ is a **cumulative distribution function (c.d.f.)** for some probability P if and only if F satisfies the following three conditions:

- 1 F is non-decreasing: $x_1 < x_2$ implies that $F(x_1) \leq F(x_2)$.
- 2 F is normalized:

$$\lim_{x \rightarrow -\infty} F(x) = 0 \quad \text{and} \quad \lim_{x \rightarrow +\infty} F(x) = 1.$$

- 3 F is right-continuous:

$$F(x) = \lim_{\substack{y \rightarrow x \\ y > x}} F(y).$$

Probability density function

- We define $f_X(x)$, the **probability density function (p.d.f.)** of X as:

$$f_X(x) = \lim_{h \rightarrow 0} \frac{1}{h} (F(x+h) - F(x)).$$

- The p.d.f. is the derivative of F_X , the c.d.f. :

$$f_X(x) = F'(x).$$

- We have:

$$F(x) = P(X \leq x) = \int_{\{y \leq x\}} f_X(y) dy \text{ and } \int_V f_X(y) dy = 1.$$

Remark: Discrete random variable

$$F(x) = P(X \leq x) = \sum_{\{y \in V | y \leq x\}} P(X = y).$$

C.d.f of a discrete r.v.

Exercise

Represente the distribution and the c.d.f of X , sum of the dots obtained rolling two dice.

x	2	3	4	5	6
$P(X = x)$	1/36	1/18	1/12	1/9	5/36

x	7	8	9	10	11	12
$P(X = x)$	1/6	5/36	1/9	1/12	1/18	1/36

Download R: <https://cran.r-project.org/bin/windows/base/> or use Python.

C.d.f of a discrete r.v.

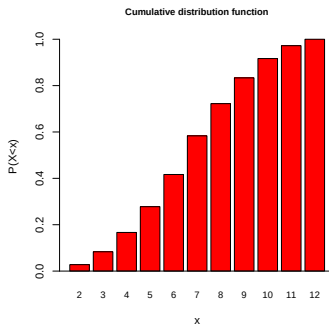
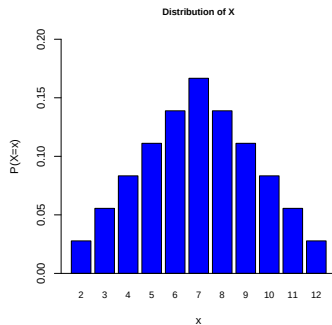
Code R

```
x<-seq(2,12,1)
Px<-c(1/36, 1/18, 1/12, 1/9, 5/36, 1/6, 5/36, 1/9, 1/12 ,1/18 ,1/36)

# Distribution of X
barplot(Px, col="blue", ylim=c(0,0.2), names.arg=c(as.character(x)),
cex.names=0.8,
xlab="x", ylab="P(X=x)", main="Distribution of X", cex.main=0.8)
abline(h=0)

# Cumulative distribution function
CdfX<-cumsum(Px)
barplot(CdfX, col="red", names.arg=c(as.character(x)), cex.names=0.8,
xlab="x", ylab="P(X<=x)", main="Cumulative distribution function",
cex.main=0.8)
abline(h=0)
```

C.d.f of a discrete r.v.



Examples of discrete r.v.'s distribution

Definition

A r.v. X is said to be a **Bernoulli's** r.v. with parameter p iff

$$P(X = x) = p^x(1 - p)^{1-x}, \quad x \in \{0, 1\}, \quad 0 < p < 1.$$

Notation: $X \sim \mathcal{B}(1, p)$.

Definition

A r.v. X follows a **binomiale** distribution with parameters (n, p) iff

$$P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}, \quad x \in \{0, \dots, n\}, \quad 0 < p < 1.$$

Notation: $X \sim \mathcal{B}(n, p)$.

Example of continuous r.v. distribution

Definition

A r.v. X has an **exponential** distribution with parameter $\theta > 0$ iff its probability density function (p.d.f) has the form:

$$f(x | \theta) = \frac{1}{\theta} \exp \left\{ -\frac{x}{\theta} \right\}, \quad x \in \mathbb{R}^+.$$

Notation: $X \sim \mathcal{E}(\theta)$

Exercise :

- 1 Check that we have a density.
- 2 Compute the cumulative distribution function.

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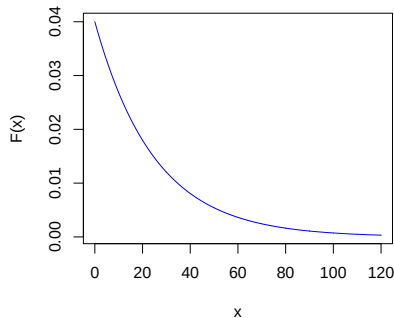
Exercise :

- 1 Check that we have a density.
- 2 Compute the cumulative distribution function.

Solution: $F(x) = 1 - e^{-x/\theta}$.

Distribution of continuous r.v.'s

P.d.f of an exponential r.v



C.d.f of an exponential r.v

